

Temperature-Sensing Diodes Selector

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Low-cost semiconductor diodes such as 1N914, 1N4148 and 1N400X can be used as temperature sensors in applications where high accuracy is not required. They can be mounted on transistors, power diodes, transformers, heat sinks, rechargeable batteries, crystals, PCB, etc to monitor their temperature.

It is highly desirable to use temperature sensors with linear temperature characteristics and the mentioned diodes are best suited for that. To use

them as temperature sensors, these diodes first need to be sorted according to their temperature coefficient. (Please refer 'Signal Diode-Based Fire Alarm' circuit idea published in February 2013 issue to understand how these diodes can be used as temperature sensors.) This circuit can help quickly sort different diodes based on their temperature coefficient.

Circuit and working

The circuit diagram of the device for selection of temperature-sensing diodes is shown in Fig. 1. The circuit is



provides the 18V power supply for the diodes (connected at CON1 through CON12) to be tested. The linear regulator IC2 provides the 9V power supply for the digital voltmeter (DVM) used for measuring the voltage drop across diodes.

Utilising this device, it is possible to test the diodes at around 0.1mA of forward current. The current for their PN junctions is provided through individual resistors R1 through R12. The value of a resistor is much higher than the resistance of the tested diode's junctions, so changing the diodes will not change the forward current significantly.

The number of the connectors for the test diodes can be increased or decreased. The device does not need any adjustment or calibration to operate properly. To calculate the temperature co-efficient (TKU) of each diode and to sort the diodes accordingly, we need to measure the voltage drop across the diodes under test.

To sort the diodes, ensure that all of them are connected in the circuit and the voltage drop (say U1) across each diode is measured one by one at the same temperature, say

25°C. The voltages can be read at TP3 through TP14 with respect to TP0 using the DVM as shown in Fig. 1. Now change the temperature to, say, 40°C, and take the measurements (say U2) again. We can use an electrical equipment, such as a heater, to produce the second temperature. So now the temperature coefficient of the PN junction

built around step-down transformer X1, voltage regulator 7818 (IC1), voltage regulator 7809 (IC2) and three diodes 1N4001 (D1 through D3). The mains supply is stepped down to 21V, 250mA using transformer X1. Diode D2 is sufficient to rectify it since the required output current is typically below 20mA. The linear regulator IC1

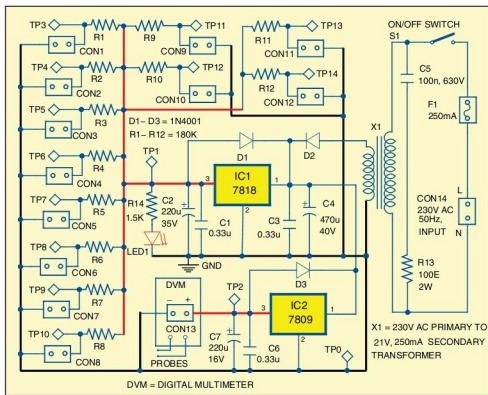


Fig. 1: Circuit diagram of the device

Test Points

Test point	Details
TP0	0V, GND
TP1	+18V
TP2	+9V for DVM
TP3 through TP14	Voltage drop across each diode